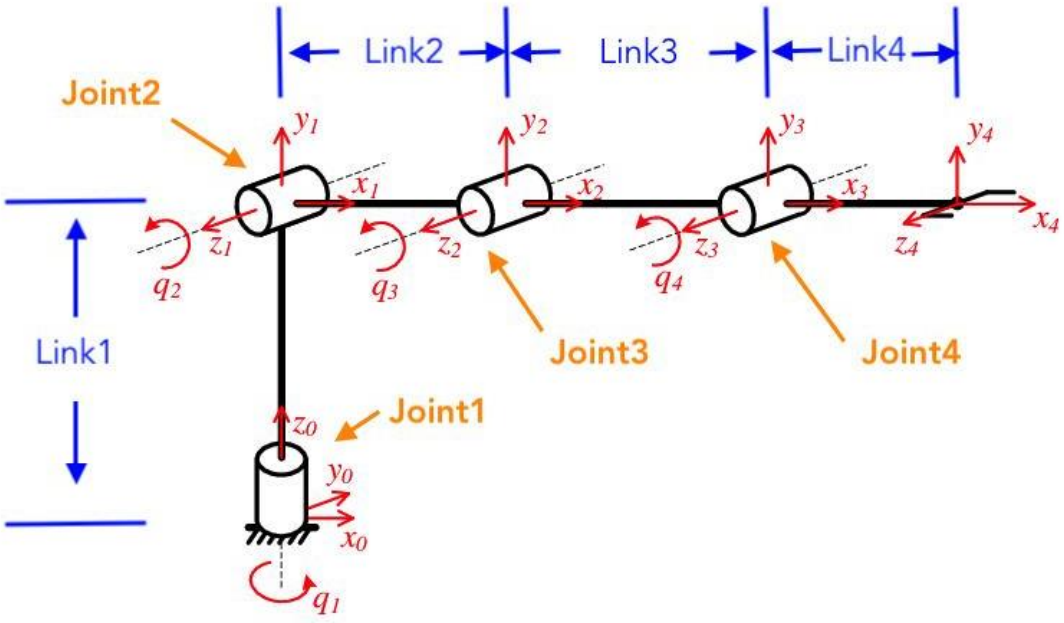
Mathematical Modelling of the 4-DOF Robotic Manipulator



Robot has four revolute joints.

Joint	Variable	Description
Base	$q1$	Rotation around vertical axis
Shoulder	$q2$	First arm rotation
Elbow	$q3$	Second arm rotation
Wrist	$q4$	Third arm rotation

The mathematical model uses these assumptions:

1. The base rotates around the vertical Z-axis.
2. The shoulder, elbow and wrist joints rotate in the same vertical plane.
3. The three-arm links are rigid.
4. Joint angles are measured in radians.
5. Link dimensions are measured in millimetres for kinematics.
6. The shoulder, elbow and wrist angles are measured from the horizontal reference, using cumulative joint angles.

For 4-DOF Robotic Manipulator Design, Modelling and Control project, will develop a mathematical model that can be used for:

1. Forward Kinematics (FK).
2. Inverse Kinematics (IK).
3. Workspace and trajectory analysis.
4. Velocity Kinematics and Jacobian modelling.
5. Dynamic modelling using the Euler–Lagrange method.
6. PID-based joint control in MATLAB/Simulink.

will use the dimensions of existing SolidWorks robot model so that the mathematical equations, MATLAB code and mechanical model remain consistent.

1. Geometrical parameters

Based on your existing robot design:

Parameter	Description	Dimension
R_0	Base radial offset	90 mm
H_0	Base vertical height	288 mm
L_1	Upper arm length	350 mm
L_2	Forearm length	335 mm
L_3	Wrist/end-effector length	225 mm

For mathematical calculations, we will use metres:

$$R_0 = 0.09m$$

$$H_0 = 0.288m$$

$$L_1 = 0.350m$$

$$L_2 = 0.335m$$

$$L_3 = 0.225m$$

Joint angle convention: All joint angles are measured in radians, and the positive shoulder, elbow and wrist angles are measured counter clockwise in the vertical robot plane.

2. Forward Kinematics

Forward kinematics determines the position and orientation of the end effector from the four joint angles.

The input vector is:

$$q = \begin{bmatrix} q1 \\ q2 \\ q3 \\ q4 \end{bmatrix}$$

The output vector is:

$$X = \begin{bmatrix} x \\ y \\ z \\ phi \end{bmatrix}$$

Where:

- x = end-effector X-coordinate.
- y = end-effector Y-coordinate.
- z = end-effector Z-coordinate.
- phi = end-effector pitch angle.

2.1 End-effector orientation

The end-effector pitch is the sum of the three rotations in the vertical plane:

$$phi = q_2 + q_3 + q_4$$

Therefore:

$$q_4 = phi - q_2 - q_3$$

2.2 Radial position

The horizontal radial distance from the robot's central vertical axis is:

$$r = R_0 + L_1 \cos(q_2) + L_2 cos(q_2 + q_3) + L_3 cos(phi)$$

2.3 X-coordinate

Because the first joint rotates the robot around its vertical axis:

$$x = r \cos q_1$$

2.4 Y-coordinate

$$y = r \sin q_1$$

2.5 Z-coordinate

The vertical position is:

$$z = H_0 + L_1 sin\, q_2 + L_2 sin(q_2 + q_3) + L_3 \sin(phi)$$

2.6 Complete forward kinematics model

Combining all four equations:

$$x = [R_0 + L_1 cos\, q_2 + L_2 cos(q_2 + q_3) + L_3 cos(phi)]\, cos\, q_1$$

$$y = [R_0 + L_1 cos\, q_2 + L_2 cos(q_2 + q_3) + L_3 \cos(phi)]\, sin\, q_1$$

$$z = H_0 + L_1 sin\, q_2 + L_2 sin(q_2 + q_3) + L_3 \sin(phi)$$

$$phi = q_2 + q_3 + q_4$$

This is the main forward kinematics model that implement in MATLAB.

3. Inverse Kinematics

Inverse kinematics calculates the required joint angles to reach a specified end-effector position and orientation.

The desired input is:

$$X_d = \begin{bmatrix} x_d \\ y_d \\ z_d \\ phi_d \end{bmatrix}$$

The required output is:

$$q = \begin{bmatrix} q1 \\ q2 \\ q3 \\ q4 \end{bmatrix}$$

3.1 Base rotation

The first joint angle is:

$$q_1 = \ atan2(y_d, x_d)$$

3.2 Calculate radial distance

$$r_d = \sqrt{x_d^2 + y_d^2}$$

Remove the base radial offset and wrist length:

$$r_w = r_d - R_0 - L_3 \cos\ (phi_d)$$

3.3 Calculate vertical wrist position

$$z_w = z_d - H_0 - L_3 \sin\ (phi_d)$$

3.4 Calculate elbow angle

Using the cosine rule:

$$D = \frac{r_w^2 + z_w^2 - L_1^2 - L_2^2}{2L_1L_2}$$

The elbow angle is:

$$q_3 = atan2\ (\pm\sqrt{1 - D^2}, D)$$

The positive and negative solutions represent different elbow configurations.

For a feasible solution:

$$|D| \leq 1$$

3.5 Calculate shoulder angle

$$q_2 = atan2(z_w, r_w)\ -\ atan2\ (L_2 sin\ q_3, L_1 + L_2 cos\ q_3)$$

3.6 Calculate wrist angle

$$q_4 = phi_d - q_2 - q_3$$

These equations provide the analytical inverse kinematics model for your robot.

4. Dynamic Modelling using the Euler–Lagrange Method

Forward and inverse kinematics describe the robot's geometry and movement. Dynamic modelling describes the forces and torques required to produce that movement.

This project uses the Euler–Lagrange formulation.

The general equation is:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathrm{q}}_{\mathrm{i}}}\right) - \frac{\partial L}{\partial q_i} = \tau_i$$

Where,

- L = Lagrangian.
- q_i= joint position.
- $\dot{\mathrm{q}}_{\mathrm{i}}$ = joint velocity.
- τ_i = applied joint torque.

4.1 Lagrangian formulation

The Lagrangian is the difference between kinetic and potential energy:

$$L = T - V$$

Where:

- T = total kinetic energy.
- V = total gravitational potential energy.

4.2 Kinetic energy

The total kinetic energy is:

$$T = \sum_{i=1}^{n} \left[\frac{1}{2} m_i \, v_{c_i}^T v_{c_i} + \frac{1}{2} \omega_i^T \, I_i \omega_i\right]$$

Where:

- m_i = mass of link i.
- v_{c_i} = linear velocity of its centre of mass.
- ω_i = angular velocity.
- I_i = rotational inertia tensor about its centre of mass.

For the four-joint manipulator, will calculate the kinetic energy of each moving link and combine them.

4.3 Potential energy

The gravitational potential energy is:

$$V = \sum_{i=1}^{n} m_i \, g \, h_i$$

Where:

- $g = 9.81 m/s^2$
- h_i = vertical position of the centre of mass of link i.

The base rotation does not directly change gravitational potential energy in an ideal vertical-axis configuration.

4.4 Complete robot dynamic equation

After applying the Euler–Lagrange equations to all four joints, the robot dynamics can be expressed as:

$$M(q)ddot\ q + C(q, dot\ q)dot\ q + \ G(q) + F(dot\ q) = \tau$$

Where:

Term	Meaning
$M(q)$	4 × 4 inertia/mass matrix
$C(q, dot\ q)$	Coriolis and centrifugal effects
$(G(q)$	Gravitational torque vector
$F(dot\ q)$	Joint friction torque vector
τ	Applied joint torque vector

The torque vector is:

$$\tau = \begin{bmatrix} \tau 1 \\ \tau 2 \\ \tau 3 \\ \tau 4 \end{bmatrix}$$

The acceleration equation becomes:

$$ddot\ q = M^{-1}(q)[\tau + C(q, dot\ q)dot\ q + \ G(q) + F(dot\ q)$$

This is the fundamental dynamic model that will use for the MATLAB simulation and PID controller development.

Important: The exact numerical entries of M, C, and G) cannot be calculated from link lengths alone. We need the mass, centre-of-mass position and inertia tensor of each moving component from your SolidWorks model.

5. Joint-space equations of motion

The four joint equations can be represented individually as:

$$\tau_1 = \sum_{j=1}^{n} M_{1j}ddot\ q_j + \ h_1(q, dot\ q) + G_1 + F_1$$

$$\tau_2 = \sum_{j=1}^{n} M_{2j}ddot\ q_j + \ h_2(q, dot\ q) + G_2 + F_2$$

$$\tau_3 = \sum_{j=1}^{n} M_{3j}ddot\ q_j + \ h_3(q, dot\ q) + G_3 + F_3$$

$$\tau_4 = \sum_{j=1}^{n} M_{4j}ddot\ q_j + \ h_4(q, dot\ q) + G_4 + F_4$$

Here, h_i represents the combined Coriolis and centrifugal contributions for joint i.

These equations will calculate the torque requirements for each motor.

6. Trajectory generation

For your robotic manipulator, we can use a fifth-order polynomial trajectory to generate smooth joint movements.

The general equation is:

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

For a movement starting at q_0 and ending at q_f, with zero initial and final velocity and acceleration:

$$q(t) = q_0 + (q_f - q_0)\,[\,10s^3 - 15s^4 + 6s^5]$$

Where:

$$s = \frac{t}{T}$$

and T is the movement duration.

This will be useful when implementing waypoint movements and drawing paths using your four-joint robot.